Supplementary Information

WestQuant Memory Discovery Lab: An Evidence-Aware Framework from Experimental Data to Quantum-Memory Simulation

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This Supplement describes the executed software and data analyses accompanying the manuscript. Experimental reanalysis, internal diagnostics, synthetic controls and numerical verification are identified separately. No new hardware experiment, independently verified new-material lifetime, or machine-learning training result is reported. Tables are generated from the archived JSON outputs where applicable.

S1 Executable interfaces and conventions

The package `memorylab` provides candidate/source records, a catalog, memory profiles, model dispatch, quantum channels, deterministic T_2 assessment and network export. Source identities are content hashes with source metadata. The base evaluated source is version 0.6.0; its archive identity is recorded in the reproduction section. The additional system evaluation package is version 0.1.0 and retains a separate source fingerprint.

Component	Input contract and interpretation
Candidate	Composition, integer charge, electronic spin $2S$, supplied geometry, optional host and site-resolved isotope substitutions. Electron/spin parity is checked; stability is not established.
Observation	Quantity, unit, conditions, source location and evidence kind. Unknown units and unresolved observable definitions remain unresolved.
Reviewed reference	Material identity, coherence definition, positive seconds, reported uncertainty convention, conditions and unresolved conditions. Matching is conditional on reported information; no reference-query interpolation.
Memory profile	Family, logical states, writing/storage/reading temperatures and durations, explicit relaxation/dephasing definitions, IO acceptance and optional independently specified bath/table coherence.
Exact bath	Nuclear I, Zeeman vectors and coupling tensors in Hz; dimension bounded; time in seconds; ideal pulse sequence; explicit Gibbs or maximally mixed initialization; selected protocol retained.
PyCCE	Conventional CCE pure coherence; backend milliseconds, kHz and gauss; explicit conversion from the common SI interface. Density elements from generalized CCE are not silently substituted.
Electronic adapters	Molecular PySCF geometry in angstrom, charge and $2S$; external coupling imports require units, site mapping and tensor-frame alignment. Periodic crystals are not molecular PySCF inputs.
Optical propagation	Separate MaxwellBloch multi-level recipe. Propagation fields do not by themselves identify a single-photon instrument.
Channel	Conditional logical CPTP map and state-independent acceptance; full qubit-to-qutrit erasure map; unnormalized column-vectorized Choi matrix.
Network device	Explicit age samples, exact requested temperature/protocol, retrieval from original input once. Missing age or unsupported intervention is an error.

Table S1: Common interface conventions and inference boundaries.

The implemented family labels are atomic ensemble, trapped atom/ion, rare-earth solid, defect spin, molecular spin and negative ion. Their availability is not six independent experimental validations. The negative-ion

branch requires an environment-overlap parameter; it does not deduce coherence from neutralization. Temperature ramps, state-dependent acceptance and a complete structure-to-spin–phonon-to-optical-instrument chain are outside the evaluated contract.

Channel compression retains Choi eigenvalues above 10^{-14} and rejects eigenvalues below -10^{-9}. Default CPTP checks use 10^{-9} tolerance; the QuTiP Markov comparison permits 2×10^{-8}. State-validity checks use 10^{-9}. Recorded maximum discrepancies, rather than merely boolean checks, are retained. Numerical agreement is verification of the specified equations, not evidence that omitted experimental physics is negligible.

S2 Source audit and reviewed coherence values

The original experimental sources are Zenodo records 14929997 and 21532904, associated with papers 10.1103/z6lc-qw2d and 10.1103/ftkb-pkvp. Both data records declare CC BY 4.0. Fourteen Teller artifacts are recorded, including processed figures and reference-source snapshots. The deterministic data branch has eleven artifacts: the PR record metadata and seven measurement files, plus three pinned molecular-repository artifacts. All seven PR measurement-file MD5 values are checked against the record metadata, and each cached artifact also has a SHA-256 digest.

The molecular data are pinned to the MTD-group molecular-qubit repository revision

`5263b1d86b39243e7405dde5bf7c58bbd8fcd677`.

Positive entries are provisional because a uniform unit, pulse definition and operating conditions have not been reviewed. There are 201 such entries across 123 source rows and 40 paper links. The repository license is not specified in the retrieved source metadata. These cached files are present for the local reproducibility bundle and excluded from default Git tracking. Their presence does not confer a license for public redistribution.

Material / source	Observable / protocol	Reported time
$^{151}Eu^{3+}$:YSO; Ortu et al.	Spin DD, XX	70 ± 2 ms
Same	Spin DD, XY4	106 ± 9 ms
Same	Spin DD, XY8	154 ± 11 ms
Same	Spin DD, XY16	230 ± 30 ms
Same	Effective optical AFC	240 ± 30 μs
V(IV) complex 1; Lin et al.	Hahn stretched T_m	2.04 ± 0.05 μs
V(IV) complex 2; Lin et al.	Hahn stretched T_m	4.11 ± 0.03 μs
V(IV) complex 3; Lin et al.	Hahn stretched T_m	4.42 ± 0.02 μs
PL6 in 4H-SiCOI; Hu et al.	Hahn T_2	4.62 ± 1.65 μs
Same	Ramsey T_2^*	460 ± 35 ns

Table S2: Ten reviewed primary-text values. These are different definitions, not a common material ranking, and are not fresh fits to raw trials. The Eu intervals are reported 95% confidence intervals. The uncertainty conventions for the other studies have not been verified.

The Eu references use an approximately 4 K cryostat and 1.35 mT along D_1. Local RF heating, the detailed bath and sequence-specific stretch exponents remain unresolved. The optical AFC expression has exponent factor four, distinct from the spin efficiency expression. See Ortu et al., npj Quantum Information 8, 29 (2022).

The molecular references apply at 5 K in an o-terphenyl glass at X-band, approximately 9.4 GHz. Complexes 1–3 have Et_3NH, n-Bu_3NH and n-Hex_3NH counterions, respectively, with $[V(C_6H_4O_2)_3]$ core. Exact selected fields, site isotope distributions and fully reviewed concentrations are unavailable in the current reference table. See Lin et al., Chemical Science 10, 548–555 (2019). The PL6 references apply to a natural-abundance 4H-SiC-on-insulator waveguide at reported room temperature; a numeric temperature and exact coherence-measurement field remain missing. See Hu et al., Nature Communications 15, 10256 (2024). None of these ten records validates an entire optical quantum-memory channel.

S3 Teller data reanalysis and internal diagnostics

The parser handles source indices explicitly. For `4.csv`, rows encode basis and zero-based cell; the six labels are X, −X, Y, −Y, 0, 1. For `8a.csv`, the basis precedes a one-based cell index; its labels are X, −X, Y, −Y,

E, L. Original headers alone would give an incorrect interpretation. Uniform means of the processed state fidelities reproduce the released figure summaries. These numerical identity checks use neither inferred raw counts nor independent experimental replication.

The spatial holdout uses odd cells for inverse-variance calibration and even cells for comparison. The fitted mean is 0.9224361066 and approximate uncertainty 0.0038483763. Covariances are unavailable; the uncertainty widths are supplied MC quantities, not asserted to be standard errors of independent sample means.

Cell	Observed fidelity	Supplied MC width	$\lvert z\rvert$
2	0.911629	0.009759	1.0301
4	0.939911	0.006394	2.3417
6	0.923065	0.007581	0.0740
8	0.902836	0.009855	1.8526
10	0.888098	0.016270	2.0539

Table S3: Within-dataset even-cell holdout. All predictions use the odd-cell calibrated mean. Shared systematics and cross-cell covariance are unavailable.

The path-point calibration is separate: pair 5,6, a 270 ns detection window, 3 K cryostat, and 18 μs total optical age comprising 10 μs AFC rephasing and 8 μs spin storage. Its conditional mean is 0.9671968089 with supplied MC uncertainty 0.0051670280. Neither the window width nor a delay in an ambiguous figure is substituted for the total optical age. A query at another storage time is rejected; no T_1 or T_2 is supplied at this point.

The acceptance model multiplies 0.16 AFC efficiency, 0.42 two-way optical transfer, 0.50 demultiplexing and 0.42 temporal acceptance. The first three reported component uncertainty widths are 0.01, 0.02 and 0.04. Under independent errors and exact last factor,

$$\sigma_\eta = \eta\sqrt{(0.01/0.16)^2 + (0.02/0.42)^2 + (0.04/0.50)^2} = 0.0015824218.$$

Temporal-acceptance uncertainty, cross-component covariance, fiber/filter/detector losses and false heralds are not included. Network Monte Carlo intervals condition on the nominal acceptance.

Readout-window diagnostics calibrate on 80 widths at or below 600 ns and assess 91 widths at or above 800 ns; 40 intermediate widths are unused. Three signal-plus-background fits have maximum pointwise discrepancies 1.728745, 2.638455 and 0.722965 for pairs 1,2; 5,6; and 9,10. Their corresponding constant-fidelity discrepancies are 5.547306, 5.279383 and 4.118887. The pair 5,6 fit reaches signal-fidelity upper bound one. Window measurements share nested counts and lack covariance, so these screens are not calibrated global hypothesis tests or T_2 estimates.

S4 Mean fidelity does not uniquely determine a channel

This is a constructive mathematical example, not a second fit to all released experimental states. For the same six-state mean F_0, define $p = 2(1 - F_0)$ and $q = 3p/4$. Depolarizing Kraus operators are

$$K_0 = \sqrt{1 - 3p/4}\, I, \quad K_x = \sqrt{p/4}\, X, \quad K_y = \sqrt{p/4}\, Y, \quad K_z = \sqrt{p/4}\, Z.$$

Phase-flip operators are $M_0 = \sqrt{1-q}I$ and $M_1 = \sqrt{q}Z$. The latter leaves the two Z-axis states unchanged and gives fidelity $1 - q$ for each of the four transverse states. Its six-state mean is $1 - 2q/3 = 1 - p/2 = F_0$.

For identical independent maps on the two halves of a Bell pair,

$$\begin{aligned} F_B^{\mathrm{dep}} &= 1 - \tfrac{3}{2}p + \tfrac{3}{4}p^2, \\ F_B^{\mathrm{phase}} &= (1-q)^2 + q^2 = 1 - \tfrac{3}{2}p + \tfrac{9}{8}p^2, \\ F_B^{\mathrm{phase}} - F_B^{\mathrm{dep}} &= \tfrac{3}{8}p^2. \end{aligned}$$

At $F_0 = 0.9671968088636844$, $p = 0.06560638227263116$ and $q = 0.04920478670447337$. Direct Kraus propagation gives $F_B^{\mathrm{dep}} = 0.9048185746372300$ and $F_B^{\mathrm{phase}} = 0.9064326486603184$, a difference of 0.0016140740230883. The matrices and values are regenerated by `scripts/make_figures.py`. More extensive tomography and a conditioning model are needed to discriminate experimentally compatible channels.

S5 PR2026 curve diagnostics and negative outcomes

Four released classical-efficiency curves have 15, 70, 70 and 70 observations for no RF, XX, XY4 and XY16. Efficiencies and their supplied SEM quantities are converted consistently; reported total delays are converted from microseconds to seconds. The fit factor is explicitly one, defining an apparent retrieval-decay scale. No result is converted to homogeneous spin T_2.

The Gaussian model is $b + A\exp[-(t/\tau)^\beta]$, with exponential $\beta = 1$ or stretched $0.25 \le \beta \le 4$. The implementation uses three starts and bounded log-τ. Every fourth sorted unique time group, using zero-based group indices congruent to three modulo four, is withheld. Thirty parametric bootstrap draws are requested per full-data fit. Status includes the full-data approximate Gaussian goodness diagnostic, holdout residuals, boundaries, observed-time range, reviewed observable mapping and bootstrap stability.

Sequence	Model	τ (ms)	Held RMSE	Approx. p	Status
XX	Exp.	0.69686	3.582	5.8e-90	Mismatch
XX	Stretch.	0.0129498	2.853	1.16e-41	Mismatch
XY16	Exp.	348.25	1.983	2.42e-87	Mismatch
XY16	Stretch.	2.39489	1.417	8.34e-59	Mismatch
XY4	Exp.	1.16409	3.786	$< 10^{-300}$ (underflow)	Mismatch
XY4	Stretch.	0.0954074	3.434	$< 10^{-300}$ (underflow)	Mismatch
none	Exp.	0.0188799	0.622	0.995	Inconclusive
none	Stretch.	0.0288762	0.468	0.998	Inconclusive

Table S4: Apparent classical retrieval scales and diagnostics; none identifies physical spin T_2. Extremely small probabilities depend on the Gaussian independence model.

Sequence	Model	β	Requested	Successful
XX	Exp.	1	30	30
XX	Stretch.	0.25	30	30
XY16	Exp.	1	30	15
XY16	Stretch.	3.2387	30	30
XY4	Exp.	1	30	30
XY4	Stretch.	0.25	30	30
none	Exp.	1	30	30
none	Stretch.	2.0866	30	28

Table S5: Requested and accepted parametric-bootstrap draws. Additional issue flags remain in the full JSON outputs.

All six RF fits are model mismatches. The no-RF exponential scale lies outside the observed window; the no-RF stretched bootstrap yields only 28 of 30 accepted draws and is inconclusive under the implemented requirement. A zero stored p is numerical underflow, not an exact zero probability. Tiny Gaussian probabilities are conditional on supplied uncertainties and independence; unmodeled correlation/systematics limit the physical interpretation. The source experiment already reports nonexponential behavior. Thus the retained negative results diagnose the tested envelopes, rather than demonstrating that the experimental memory itself fails.

The separate heralded-photon arrays are not qubit process tomography. The source value `st`=100 μs and the article's 100.9 μs value are retained as an unresolved clock mapping. The paper's nominal spin coherence quoted from other work is not a new lifetime measured from these arrays. Neither $g^{(2)}$ nor extended classical storage is substituted for time-resolved qubit fidelity.

S6 Deterministic and synthetic verification

The Markov case supplies $\Gamma_1 = 10$ and $\Gamma_\phi = 20\ \mathrm{s}^{-1}$ at hypothetical 4.2 K conditions, giving 40 ms. Rate-box corners use $\Gamma_1 \in [8, 12]$ and $\Gamma_\phi \in [18, 22]\ \mathrm{s}^{-1}$. Their range [35.714286,45.454545] ms is deterministic parameter uncertainty; its $-10.7143\%/+13.6364\%$ deviations are not prediction-error margins.

The finite-bath case has one ^{13}C spin $I = 1/2$, logical projections $m = \pm 1/2$, laboratory Zeeman $\omega_z = 214163.9829$ Hz and hyperfine $A_{zz} = 4000$ Hz at 4.2 K. Other couplings vanish. In the unpolarized

limit, the longitudinal model gives $L(t) = \cos(\pi A_{zz} t)$; the finite-temperature implementation uses its exact Gibbs state. Each of free evolution and ideal one-pulse Hahn control evaluates 101 times over 0–1 ms. Free evolution has a first sampled $1/e$ crossing bracket [90,100] μs and revivals. Hahn control swaps the commuting branches and has no sampled crossing over this finite window. These parameters are specified controls, not inferred properties of a molecular candidate.

The synthetic likelihood control uses $A = 0.8$, $b = 0$, $\tau = 1$ ms, exponential factor one and four known phases $0, \pi/2, \pi, 3\pi/2$. Ages are zero plus 20 geometric points from 10 μs to 4 ms. At every age and phase, 4000 binomial trials are generated with seed 20261010: 84 count observations and 336,000 Bernoulli trials. Group holdout retains all four phases at each withheld age together. Five ages (20 observations) are held out; 64 observations calibrate the holdout fit. The reported full-data fit uses all 84 counts. Sixty requested bootstrap draws all succeed. Recovery is 0.9796631569 ms with conditional 95% interval [0.9502149602,1.0077371898] ms, held-out Pearson RMSE 0.8632951 and approximate goodness $p = 0.9528551$. One recovered generator instance is not a coverage study.

S7 Network controls and work counts

The Teller integration control uses the actual SeQUeNCe 1.2.0 event kernel and density manager. Each attempt loads an ideal Bell state, schedules retrieval at 18 μs and applies the calibrated map once. Two nodes have nominal acceptance 0.014112, independent losses and ideal link/herald probability one. Four seeds 20261010–20261013 each have 100,000 attempts at a 1 ms interval. Counts 20, 32, 24 and 11 sum to 87. The fixed-loss expectation is 79.6594176 pairs. The pooled Wilson interval is [0.000176350196,0.000268249181]. Conditional Bell fidelity is 0.9048185746, and the maximum joint-density discrepancy is 1.11×10^{-16}. The simulated attempt clock, totaling 400 s, is not wall-clock runtime or a physical network benchmark.

The separate hypothetical profile has write/read acceptance 0.8/0.7, depolarization parameters 0.01/0.02, write/read durations 100/200 μs and constant 4.2 K stages. It exports ages 0, 100 and 500 μs; retrieval uses 500 μs, ideal link/herald factor 0.5, 2000 attempts and 1 ms interval. It produces 321 pairs and Bell fidelity 0.9559176317. Its maximum channel Choi discrepancy is 4.00×10^{-15} and joint-density discrepancy 5.55×10^{-17}. These supplied parameters have no experimental calibration in this study.

Executed/recorded unit	Count
Processed Teller state-fidelity values	90
Spatial held-out cells	5
Primary window fits / constant diagnostics	3 / 3
PR classical curve observations	225
Primary PR curve-model fits	8
Requested PR bootstrap replicates	240
Deterministic model specifications	3
Exact spin-bath time evaluations	202
Markov parameter-box corners	4
Synthetic count observations / Bernoulli trials	84 / 336,000
Synthetic requested bootstrap replicates	60
Teller network seeds / attempts	4 / 400,000
Hypothetical network attempts	2000
Original / additional evaluation tests passing	75 / 7
New hardware experiments / ML training runs	0 / 0

Table S6: Different work units are reported separately. Optimizer function evaluations, three starting points, holdout refits and bootstrap refits are not counted as eight additional primary experiments. The additional bounded local worker benchmark is reported separately below; cluster-scale performance remains untested.

S8 Additional system evaluation: complete executed protocol

Steps 1, 3 and 4 have executed results at their stated software scope. Step 2 has an executed synthetic second-family case, whereas its experimental physical objective remains blocked. The four historical selected records share evaluation-source fingerprint

`d8fc6d2ecb580e0ba1c8881ffbd542d087c493b7621cb89509948e595211ec92`.

Their common lab-source fingerprint is

`9c4490acf2e5a42e8e10007989eeb94b5e406b17f5d187ca310847581ee2edb7`.

These fingerprints identify source content, not hardware or an external scientific endorsement. Steps 2 and 3 were confirmed with the published implementation after their initial runs; the initial outputs remain archived. Table generation asserts the shared fingerprint and expected numerical/status gates without overwriting the original run records. The configuration digest and UTC timestamp of each selected record are in `evidence/system_evaluation_claim_map.json`. Generated table files are under `evaluation/paper/`.

S8.1 Step 1: independent reference equations and matched inputs

The reference `evaluation/wqeval/reference.py` imports NumPy/SciPy without importing `memorylab`. For column vectorization and Hamiltonian $G = \mathrm{diag}(0, \delta)$ in Hz, it uses

$$\mathcal{L} = -2\pi i(I \otimes G - G^{\mathsf{T}} \otimes I) + \sum_J \left[J^* \otimes J - \tfrac{1}{2} I \otimes J^\dagger J - \tfrac{1}{2}(J^\dagger J)^{\mathsf{T}} \otimes I\right].$$

The jump operators are $\sqrt{\Gamma_1(1-p_e)}\,|0\rangle\langle 1|$, $\sqrt{\Gamma_1 p_e}\,|1\rangle\langle 0|$, and $\sqrt{\Gamma_\phi/2}\,Z$. The stage superoperator is $\exp(t\mathcal{L})$. Write and read stages include their specified depolarizing maps, and storage adds the declared network-activity dephasing. The reference constructs $C = \sum_{a,b} |a\rangle\langle b| \otimes \mathcal{E}(|a\rangle\langle b|)$ by matrix-unit propagation, including an orthogonal erasure outcome for the full channel. It computes six Pauli-eigenstate outputs and their mean target fidelity independently of the lab's Kraus construction.

The four joint parameter choices (Γ_1, δ, p_e) are $(0, 0, 0)$, $(0.05, 123, 0.1)$, $(10, -500, 0.5)$ and $(100, 1000, 0.9)$, with rates in s^{-1} and detunings in Hz. Each is combined with storage temperatures 4.2, 10 and 80 K and ages $[0, 10^{-6}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}]$ s. Write/read stages remain at 4.2 K, last $10^{-4}/2 \times 10^{-4}$ s and use acceptance 0.8/0.7 and depolarization 0.01/0.02. The pure-dephasing rate is $0.01 + 0.001T + 0.001(T/10)^3$ s^{-1}, erasure rate 0.01 s^{-1}, and storage activity contributes $10 \times 0.002 = 0.02$ s^{-1}. Rate validity is declared over 1–300 K. These are chosen software profiles, not fitted material parameters. The measured Teller point supplies the thirteenth profile and its sole supported age of 18 μs at 3 K, giving $12 \times 6 + 1 = 73$ channel-age points.

The two experimental input files are `4.csv` and `8a.csv` from record 14929997:v1. Their SHA-256 hashes, original download URLs and data-license metadata are pinned in `evaluation/data/teller_sources.json`. Both parsers use the same checked bytes. The path file yields 30 values and mean 0.9510824228990328; the time-bin file yields 60 values and mean 0.9140479475068066. The discrepancy is zero for every processed value. The comparison threshold is 10^{-8} for the full Choi matrix, six states, average fidelity, acceptance and processed values. This threshold is numerical, not an experimental accuracy margin.

Both timing arms are warmed up before five repeats, with their order randomized by seed 20261010. All 73 channel constructions are included in each repeat. `construction_timing.csv` retains every measured duration and actual order. The arms do not perform equal amounts of metadata, export, validation or metric-construction work. File IO, source retrieval, network events, fits and electronic structure are excluded. Native thread environment variables are recorded but were not constrained for step 1. The medians cannot be interpreted as a fair end-to-end speedup or engineering-effort reduction. `related_tools_audit.json` distinguishes executed comparison from documentary context; AiiDA, PySCF, PyCCE, QuTiP and SeQUeNCe were not separately timed here. Existing AiiDA provenance and caching are not presented as new contributions.

S8.2 Step 2: synthetic control, curve gate and physical stop

The model/curve comparison requires the observable `complex_pure_dephasing_coherence`, explicit exclusion of population relaxation, source hashes, curve reviewer, normalization and observable mapping. A measured case additionally requires reviewed model-parameter provenance, a nonhypothetical model and exact equality of full declared experimental conditions between curve and model. The fixed-model runner

accepts `exact_spin_bath` or optional `pycce`; it fits zero parameters. The initial curve value must be unity, magnitudes must lie in the unit disk and a strictly increasing finite grid must start at zero. No silent clipping of noisy or unnormalized measurements is provided.

The executed control has one ^{13}C nuclear spin $I = 1/2$, central logical projections $\pm 1/2$, laboratory $A_{zz} = 4000$ Hz and $\nu_z = 214163.9829$ Hz, zero other couplings, 4.2 K and zero pulses. Forty-one equally spaced ages span 0–1 ms. An independent closed-form Gibbs-initialized FID produces the curve; the finite Hamiltonian solver supplies the comparison. Every complex residual is divided by the chosen scale 0.01, and the maximum must be at most three. The observed maximum is $3.7192473 \times 10^{-12}$. This chosen scale is not an experimental SEM, and the gate has no calibrated statistical false-positive rate. The 75–100 μs crossing bracket differs from the earlier 101-point control's 90–100 μs bracket solely because the grids differ; both are sampled numerical brackets and both detect revivals.

All write/store/read temperatures are hypothetically 4.2 K. The supplied interface has acceptance 0.8/0.7, depolarization 0.01/0.02 and durations 100/200 μs, with the independent rate models retained from the example. Export requires `-allow-hypothetical`. At 500 μs, the provider-neutral two-node workload has 10,000 trials, herald probability 0.1, 1 ms interval, Bell-fidelity threshold 0.8 and seed 20261010. It yields 1020 heralds and 313 retrieved/quality-qualified pairs. Conditional Bell fidelity is 0.9559176317211326; the retrieved-fraction Wilson interval is [0.0280630600,0.0348969001]. The simulated qualified rate is 31.3 s^{-1} under these supplied assumptions, not measured network performance. This workload is not the earlier 2000-trial SeQUeNCe control.

The separate pending descriptor targets a molecular reference at DOI 10.1039/C8SC04122A. Running it returns `blocked_missing_reviewed_data` and exit code two, with no physical validation and no network export. Required inputs are a reviewed normalized curve with measurement uncertainties, fixed Hamiltonian and tensor frames, field/temperature/isotopes/pulses, separation of relaxation from pure dephasing, and independent optical-interface review. Reported molecular phase-memory time alone does not satisfy these requirements. Thus no independently parameterized second-family physical validation or new-material lifetime is supplied by this original step-2 pathway. The subsequent Bradley observable case is described separately below.

S8.3 Step 3: individual fault interventions

The suite creates local copies of inputs and explicitly unsafe bypasses without modifying production guards. Each invalid case has a valid positive control that must succeed. Rejection alone would be insufficient if every valid input also failed. All seven positive controls and all seven guarded rejections pass; the deliberate bypasses accept all seven cases.

Constructed fault	Guarded result	Accepted unsafe-bypass consequence
Operating-point age extrapolation	Blocked	10× unsupported age
Operating-point temperature extrapolation	Blocked	1 K unsupported shift
Total coherence rate reused as pure dephasing	Blocked	16.667% conditional T_2 error
Unreviewed noise independence	Blocked	One unsupported declaration
Hypothetical channel without opt-in	Blocked	Two hypothetical age samples
Absolute-age map applied incrementally	Blocked	0.097247 maximum element error
Modified source bytes	Blocked	1000× forged source value

Table S7: Step 3: all seven valid positive controls pass; all invalid cases are rejected. Consequences have distinct units and are not aggregated into an accuracy score.

The age bypass changes the declared sole point age before evaluating a tenfold longer age, thereby illustrating an unsupported extrapolation rather than measuring its physical error. The temperature bypass changes the declared point temperature by 1 K, again with no experimentally known extrapolation error. Noise independence is counted as an unsupported declaration; evidence promotion counts two age samples. The rate-definition example has $\Gamma_1 = 10$ s^{-1} and correct independent $\Gamma_\phi = 20$ s^{-1}, giving 40 ms. Treating total coherence rate 25 s^{-1} as pure dephasing gives 33.333 ms instead. The modified-source case changes a chosen value from 0.001 to 1 s; it exercises the new evaluation source contract, whereas the other six cases exercise existing lab guards.

In the incremental-application control, the 1 ms complete channel includes write/read depolarization as well as storage; applying it twice produces maximum density-element discrepancy 0.0972467329 relative to

applying it once. A separate units control converts 1 ms correctly to 0.001 s and contrasts it with an omitted conversion to 1 s; its maximum element discrepancy is 0.1322008762. These consequences have different units and scientific meanings. Neither their average nor seven out of seven represents a general material-prediction accuracy or real-world researcher error rate. Equivalent guards may be implemented in other frameworks.

S8.4 Step 4: workload, timing scope and robustness

The native runner workload has Markov and one-spin exact-bath models, detunings $[0, 123, 500]$ Hz, two replicates and six ages $[0, 10^{-4}, 5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}, 5 \times 10^{-3}]$ s. Replicates have distinct deterministic seed streams; they are not independent hardware measurements or stochastic bath realizations in this deterministic workload. The Cartesian count is $2 \times 3 \times 2 = 12$ jobs per study. Master seed is 20261010. Three repeats independently randomize worker order among 1, 2 and 4. Worker count is excluded from physics identity, allowing a direct consistency check across scheduling choices.

Each cold study runs in a separate output directory and an isolated measurement process. OMP, MKL, OpenBLAS and NumExpr thread limits are one in those benchmark subprocesses. Per-job timeout is 60 s, driver timeout 600 s and max-run budget 24. The measured interval includes process startup, actual channel construction and artifact IO. The warm interval reuses terminal-result checkpoints; it is not electronic-structure memoization. Nine cold studies complete 108 jobs, and nine warm passes reuse 108 results. The scientific digest, excluding durations and output paths but including configuration identities and channel data, is identical across all nine studies.

Workers	Cold median [min,max] (s)	Resume median (s)	Median jobs/s
1	3.8674 [3.8089, 3.9101]	0.01814	3.103
2	1.9781 [1.9165, 2.0665]	0.01741	6.066
4	1.1782 [1.1299, 1.2286]	0.01817	10.185

Table S8: Step 4: three local cold studies and three checkpoint-reuse passes per worker count, with 12 jobs per study. Bounds are observed minima/maxima, not confidence intervals.

The reported bounds are observed ranges from three repeats, not confidence intervals. Separate resource measurements use process usage counters: parent maximum RSS is 34.578125–37.078125 MiB and the largest terminated child's maximum is 63.238281–66.414063 MiB. The metric is not peak total memory across concurrent workers. Child CPU seconds and each measurement's raw counters remain in the retained summary. Performance applies to this Linux environment and small supplied models, not general CCE, electronic structure, all elements or cluster-scale workloads.

Three additional robustness probes are distinct from the 108 cold jobs. A profile without write temperature produces `missing_data`. A real worker process with a 10^{-6} s timeout produces `timeout`; this deadline intentionally tests termination and is not a solve-time estimate. A precompleted valid job followed by the full study completes all 12 jobs with exactly one checkpoint reused. Thus robustness adds one missing-data worker, one timeout worker and 12 successful calculations across the partial/restarted study. No forced-kill during an atomic write, power-loss, corruption or concurrent cache-writer experiment was performed. Failed terminal checkpoints are preserved as failures, not retried or counted as successful physics results.

S8.5 Core fresh-checkout CI and traceable work units

GitHub Actions run 38051481969 at commit `de52b32f0c34a05cf4f6a0c01bca6ee087e6cb67` uses Ubuntu 24.04 and Python 3.12.15. It installs the core editable package with NumPy 2.5.3 and SciPy 1.18.1, passes seven evaluation tests, automatically retrieves/checks both pinned experimental files, and reruns the matched comparison. Its maximum full-Choi discrepancy is $1.3214015 \times 10^{-14}$ versus the 10^{-8} gate. `evidence/core_ci_job.log` and `core_ci_verification.json` preserve the log and summary; the log hash is checked. This core CI result is additional environment evidence, not full optional-backend reproduction or colleague-led independent scientific review. CSV bytes are preserved and automatic Git line-ending conversion is disabled to protect source hashes.

At that historical system-evaluation snapshot, the original 75 tests plus seven additional evaluation tests constituted 82 distinct local passing tests. Repeating controls, CI tests, numerical ages or Monte Carlo trials does not increase that count. Work units remain separate: 73 channel-age comparisons, 90 processed data values, 41 synthetic coherence ages, 10,000 provider-neutral network trials, seven selected invalid contracts

with seven positive controls, 108 cold worker jobs and 108 checkpoint reuses. No new hardware experiment or ML training run is counted. Steps 5 and 6 remain the author's and colleagues' full independent reproduction and scientific/release review tasks.

S9 Reproduction and claim traceability

The source archive used for this study has SHA-256

`c64b94a573def9d999b65be7201947e9c`
`2ce4e496aaf201650d7c1e3d44733aa`.

The code and article are now public at https://github.com/VesterlundCoder/memory-discover-lab. The core CI release commit is recorded above; the original private source-archive hash remains a separate provenance identifier. No software DOI or peer-review acceptance is asserted. The repository root contains the manuscript, figures, tables, evidence, scripts and evaluated source in `lab/`. `REPRODUCIBILITY.md` supplies constrained Python 3.12 setup and LaTeX requirements. The current public clone uses default offline licensed numerical replay with `scripts/reproduce.py`; `-online` explicitly selects wider source acquisition. Section S10 defines the excluded molecular acquisition and unbundled paper-byte checks. Fresh replay outputs go into a separate directory and do not replace archived article evidence.

`scripts/verify_results.py` checks the reanalysis values, count controls, model statuses and simulated counts, and emits a claim-to-result JSON map. These checks protect the reporting pipeline; they are not extra experimental measurements. Every headline claim links to an executed result or log; the later manuscript update has a separate generated claim map described below. Detailed fit outputs retain status, seed, bootstrap success, conditions and limitations. Files predating this article remain under `lab/` as historical source outputs and are not all fresh experimental validations. The `evidence/` directory is the article's selected result snapshot.

The system-paper tables and claim map are regenerated with `python scripts/document_system_evaluation.py`; the numerical result files are left intact. The complete manuscript can be built using `make pdf`. Numerical reproduction does not require LaTeX; PDF compilation does not require running the simulator. Figures require Matplotlib. No ML libraries are needed. The original archived software test logs record Python 3.12.14, NumPy 2.5.3, SciPy 1.18.1, QuTiP 5.3.1, PySCF 2.14.0, PyCCE 1.1.1 and SeQUeNCe 1.2.0. Optional-adapter test execution does not replace application-specific physical validation.

The four evaluation commands use fresh `-out` directories: `python scripts/evaluate_system.py step1`, `step3` and `step4`, plus `step2 -config evaluation/data/software_control/case.json -allow-hypothetical`. The pending physical descriptor is the default for `step2` and returns exit two. Full commands and source contracts are in `evaluation/README.md`; manuscript-ready evaluation prose is under `evaluation/paper/`.

Future candidate evaluation should retain these distinctions: independently review units, definitions and conditions; anchor Hamiltonian and noise parameters; establish numerical convergence; reproduce time-resolved optical fidelity, success and noise; and assess any proposed improvement under matched operating conditions. The present study supplies the workflow and controls for such work without claiming those future outcomes.

S10 Clean-clone reproduction repair

The review of commit `de52b32` identified clean-clone acquisition, test and figure-generation failures. The later documentation commit retained confirmations, but the executable repair is version 0.6.1. It changes source acquisition, test fixtures and reporting/reproduction interfaces without changing the microscopic equations. Historical source identities and results are retained separately. Current outputs are in `evaluation/results/repaired/`; the repair manifest, execution/test logs, four comparison reports and paper replay claim map are under `evidence/repair/`.

S10.1 Inputs, execution mode and scientific scope

The tracked bundle contains ten Teller CSV files and seven PR2026 OUT/NPZ files. Their unchanged bytes are verified against SHA256, and Teller replay additionally checks the original reviewed MD5 locks. The two data records declare CC BY 4.0; creator names, DOI, URLs, fixed revisions and license links are retained in `data/experimental/ATTRIBUTION.md` and `manifest.json`. Unused `3a.csv`, primary-paper PDFs and the unlicensed molecular raw collection are not bundled.

The default `scripts/reproduce.py` uses these licensed numerical inputs and retained reviewed paper transcriptions. It writes a fresh tree, refuses a nonempty output directory and verifies required dependencies/data before execution. It reexecutes the Teller consistency, holdout, readout-window, channel, QuTiP and four SeQUeNCe analyses; PR curve fits; reference-table, exact-bath and count controls; and the hypothetical network workload. It then regenerates figures/tables and runs ten reporting checks. Archived article evidence is not replaced. Records explicitly distinguish verified measurement bytes from unbundled reviewed paper references.

Offline replay does not reacquire the 201-entry molecular source collection or reverify unbundled paper bytes. Its freshly acquired molecular-observation count is zero, and its reporting mode checks that exclusion. The historical count remains attributed to the original archive. `-online` explicitly selects the wider original acquisition adapters and requires the external endpoints. That path was not rerun in the denied-network repair validation; denied endpoints and changed bytes remain failures.

S10.2 Clean clone and installed environments

An author-operated local candidate Git repository was cloned into an empty tree, with no ignored source cache, processed-data tree, molecular source-lock file or preceding synthetic-demo output. Isolated core and full virtual environments were installed on Linux/Python 3.12.14. After installation, a test-only Python socket/DNS shim denied external destinations and allowed loopback for local HTTP download tests. This verifies absence of external retrieval during the tested commands; it is not a general security sandbox. Complete installed dependency constraints cover 14 core and 131 full distributions, with the editable project installed separately. CI and Docker use the constraints, and the recommended runtime is Python 3.12. Other Python/platform combinations are not claimed tested.

The database test now supplies temporary synthetic provisional rows, while the synthetic count fixture is a tracked owned input. Optional QuTiP/SeQUeNCe checks have availability skips. The core environment runs 75 lab tests: 68 pass and seven are skipped; all 18 evaluation/regression tests pass. The isolated full environment passes 75 plus 18 tests with no skips. The eleven added regressions cover offline inputs, corruption, validation before writes, preservation of figures on missing input, metadata exclusion, changed scientific output, nested channel changes and strict source mismatch, plus three Kraus-map consistency controls described below. A skipped test is not a passing solver validation.

S10.3 Current execution identity and numerical comparisons

Two complete system passes share evaluation-source SHA256

`254b1bbb8a1d3c95d28b63e69d40c3a7afcbc2aa76b4b14a2bf8a727c9ead1e9`.

Their lab-source SHA256 is

`99b0ef35caf1791653ce2b9c029514b219340ed58198194e291bccc4095e62f`.

These content fingerprints identify the archived repair revision. Later lab/evaluation extensions have their own source fingerprints; they are not expected to equal these historical identities. The candidate local

Git identity, UTC times, full configuration and installed environment remain in original outputs; they are not relabeled as a remote release.

Each pass performs 73 channel-age comparisons and 90 processed fidelity comparisons. The full-Choi discrepancy remains $1.3103013 \times 10^{-14}$. The 41-point synthetic step-2 control and hypothetical network gate pass, with physical validation false. Seven faults are rejected and all seven valid positive controls pass. Each local benchmark pass completes 108 cold jobs and 108 terminal-checkpoint reuses, giving 216 of each over the two passes; robustness probes are counted separately. The physical descriptor returns exit 2 with `blocked_missing_reviewed_data` in both passes. The paper replay gives 87 retrieved pairs in 400,000 Teller attempts, six mismatching and two inconclusive PR fits, and 321 retrieved pairs in the separate 2000-trial hypothetical SeQUeNCe case.

`scripts/verify_evaluation.py` compares scientific fields, configuration, input hashes and seeds at 10^{-10} absolute and relative tolerance. Step 4 requires `scientific_results.json`, containing the complete channel/configuration payload with execution-derived IDs and environment removed. Source fingerprints are separately reported and optionally required to match. Timestamps, paths, timing, RSS/CPU and diagnostic path strings are excluded from numerical equality. Changed metrics/channels are rejected. Legacy step-4 summaries with source-dependent digests alone are insufficient for portable cross-version channel comparison. All four strict source/scientific comparisons pass. Runtime records remain evidence of execution, not portable physical predictions; repair wall times do not establish a performance improvement, and other author-validation tasks could be active.

The first remote repair CI run (38058979906) passed the full offline paper replay and core test suites, but rejected individual Kraus entries in the step-2 comparison. Across runners their imaginary entries differed by about 1.16×10^{-8} although the maps were equivalent. Since Kraus decompositions are nonunique, equality now uses $J = \sum_k \mathrm{vec}(K_k)\mathrm{vec}(K_k)^\dagger$, reconstructed separately for the full and conditional maps, at the unchanged 10^{-10} tolerance. The stored full Choi matrix is also checked for consistency. Basis, vectorization, physical configuration and metrics remain compared. Three additional regressions accept phase/unitary changes of Kraus representation and reject a changed conditional map or operators inconsistent with the stored Choi matrix. The negative CI result and former source identity are archived rather than omitted.

S10.4 Figures, optional imports and unexecuted scope

The figure builder validates/loads inputs before writing and stages plots before replacement. Missing-input regression returns exit 2 and preserves an existing architecture figure. PDF creation/modification timestamps are removed; two figure-only runs in the constrained environment are byte identical for all five PDFs and five PNGs. No equivalence across untested plotting/font versions is assumed. Delivered byte checksums are regenerated separately from numerical equality.

In a fresh process, direct PyCCE 1.1.1 import fails at the removed SciPy `isintlike` location. The adapter restores the same SciPy utility and NumPy Unicode dtype alias before import; adapter import and the independent one-spin analytical test pass. This compatibility route is documented. The pinned Dockerfile is updated but Docker was unavailable, so no image-build result is claimed. TeX sources were compiled and PDFs visually inspected separately. Author-operated clean clones/CI do not complete colleagues' independent reproduction or scientific/release review, and no improved material lifetime is established.

S11 Distinct-family Bradley observable reanalysis

The additional dataset is 4TU.ResearchData version 3, associated with Bradley et al., npj Quantum Information 8, 122 (2022). Its archive MD5 is `0e1fb4500688659f9dcca3123831e4f3`. The bundle retains the original archive, source analysis notebook and extracted `data_3b.json`; SHA-256 identities and CC BY 4.0 attribution are in `data/experimental/bradley2022/provenance.json`. The runner verifies that the extracted bytes match the nested source archive. Source fitted arrays are not treated as measurements.

S11.1 Conditions, Hamiltonian and measurement model

The memory is one ^{13}C spin ($I = 1/2$) near an NV centre in type-IIa CVD diamond with target 0.01% ^{13}C composition. The sample temperature is 4 K; the reviewed NV-frame field is $(B_x, B_y, B_z) = (0.3 \pm 0.1, 0.06 \pm 0.08, 46.801 \pm 0.001)$ G. Fig. 3b uses free evolution in heralded NV^0, no refocusing pulses, 1 ms ionization/recharging pulses, and NV$^-$ postselection after recapture. Nuclear-spin data are not SPAM-corrected. A site-resolved bath realization is not identified by the nominal isotope fraction.

The source gives $A_\parallel = 80 \pm 1$ Hz and $A_\perp = 271 \pm 4$ Hz measured in NV$^-$. Transferring them to NV^0 would be an additional physical assumption. NV^0 electron relaxation inferred from the same envelope is also not independent validation data. The executed comparison instead uses $H_{\text{eff}}/h = \Delta\nu I_z$, with $I_z = \text{diag}(1/2, -1/2)$ and $\Delta\nu = 79.8$ Hz. A separate NMR measurement reports 50 229.8±0.1 Hz; the RF reference is 50 150 Hz, corresponding to the source's artificial 80 Hz detuning relative to 50 230 Hz. The frequency is fixed, not fitted to held-out X.

The envelope is $E(t) = \exp[-(t/\tau)^\beta]$. With $U = \exp(-i2\pi\Delta\nu I_z t)$, the normalized channel has Kraus operators $\sqrt{(1+E)/2}\,U$ and $\sqrt{(1-E)/2}\,UZ$. The measured transverse length is $AE(t)$ and signed projection is $E(t)[c\cos(2\pi\Delta\nu t) + s\sin(2\pi\Delta\nu t)]$. The parameters A, c, s belong to preparation/readout amplitudes, not channel attenuation. Independent matrix-exponential propagation checks the channel convention, CP/TP and physical states. No noisy measured Bloch length is clipped, normalized to one or given an artificial zero-age point.

S11.2 Actual calibration, held-out diagnostics and failure

The envelope has 13 ages: nine calibration points through 80.3 ms and four held-out points at 90.3–120.3 ms. Signed X has 11 ages: five calibration points through 6.18 ms and six held-out points thereafter. The separate series have separate signal amplitudes. The split was selected after source inspection. A, τ, β fit only envelope calibration points; c, s fit only the first five X points, with envelope parameters retained. No optimizer uses the withheld values. Fits are deterministic and require no sampling seed.

The retained fit is $A = 0.53533$, $\tau = 55.247$ ms, $\beta = 1.36263$, $c = 0.48286$ and $s = 0.02318$. The original all-envelope fit reports 57±3 ms and $\beta = 1.2 \pm 0.1$; the subset scale here is not a new measured T_2 or confidence interval. Each gate compares the maximum absolute held-out residual divided by its unchanged source error with three. Envelope discrepancy 2.428 passes; X discrepancy 3.158 fails. X sensitivity at 79.7 and 79.9 Hz gives 3.2085 and 3.1070, respectively, so the failure is retained without changing the gate.

Source covariance, raw-shot likelihood and propagation of calibration uncertainty are unavailable. The positive Bloch length may also have an unmodeled noise floor. This is therefore a descriptive adequacy screen and cannot refute the original experiment's interpretation. The completed observable comparison retains `physical_validation_passed=false`, `microscopic_model_validated=false` and `network_exported=false`. It does not supply the original strict normalized-complex pure-dephasing case or optical IO review.

S12 Joint write-store-read and detection controls

The source profile is pinned by hash in `evaluation/config/validation_extension/protocol.json`. Controls override its storage description with supplied Markov rates; an unexecuted nuclear-bath noise contribution is removed from that budget. IO times are 100/200 μs for writing/reading. A separate named readout background uses a 10 μs detection window. Its declaration must be independent of signal noise and cannot reuse an existing source identity. The event model is `exclusive_signal_then_background`.

Seven instrument cases at eight ages $[0, 10^{-4}, 5\times10^{-4}, 10^{-3}, 0.005, 0.01, 0.03, 0.1]$ s give 56 complete channel-age maps and 336 state probes. Cases cover identity, full loss, background only, dephasing/loss,

thermal/background, biased background and noise dominance. Each configuration records relaxation, dephasing, erasure, equilibrium population, detuning, write/read efficiency/depolarization and background population/rates. The independently formed matrix-unit/Liouvillian Choi map and six outputs agree with the lab at 10^{-10} tolerance, with maximum discrepancy 1.443×10^{-15}. CP/TP and event probabilities are also checked. At zero acceptance the conditional fidelity is missing. A background-only map can have acceptance but has signal efficiency zero and average target overlap 1/2.

The experiment audit checks the bundled PR2026 NPZ with SHA-256

`110cbe3846f1068ea099902b3d834fad599`
`2025dd7d8f9608f49d7eb68cb951f`.

Nine source ages retain 27 central values for efficiency, noise and $g^{(2)}$, plus their errors. Every qubit-fidelity field is missing. Clock offset, herald/vacuum trial semantics and absent covariance remain unresolved. Neither Teller nor Bradley is substituted to fill these fields. `experimental_pending.json` specifies common conditions and required tomography, signal and background counts. This experiment gate is inconclusive independently of the numerical pass; no material or hardware validation follows.

S13 Hamiltonian and refinement recipes for the synthetic baths

All nuclear spins are ^{13}C with $I = 1/2$, central projections $m = \pm 1/2$, zero quadrupole terms and maximally mixed initialization. For site index $i = 0, \ldots, n-1$, define $r_i = 3.5(1.7)^i$ angstrom and $\theta_i = 0.8i$. Its laboratory position is $(r_i \cos\theta_i, r_i \sin\theta_i, 0.3r_i)$. The supplied hyperfine tensor has only its third row nonzero,

$$(A_{i,zx}, A_{i,zy}, A_{i,zz}) = (1600, 300, 2400)(3.5/r_i)^3 \text{ Hz}.$$

Every pair $i < j$ is explicitly specified as

$$J_{ij} = 120(0.25)^{i+j-1}\,\mathrm{diag}(1, 1, -2) \text{ Hz}.$$

The field is $(0, 0, 0.0002)$ T. With the retained PyCCE isotope convention $\gamma_{13C} = 6.72828532$ rad ms^{-1} G^{-1}, the exact solver's Zeeman vector is $-\gamma_{13C}\mathbf{B}\,10^7/(2\pi)$ Hz. This sign, angular-frequency conversion and field scaling are checked explicitly. Pair/hyperfine Hz convert to PyCCE kHz, time seconds to milliseconds and field tesla to gauss. Absent PyCCE pairs can otherwise acquire dipolar couplings, so a supplied zero interaction must also be explicit when used to suppress one.

S13.1 Independent references and actual counts

Four spins define the exact/reference comparisons. Matrix-exponential and independent Hermitian eigensystem propagation construct the branch unitaries and $L(t) = \mathrm{Tr}[U_0 U_1^\dagger]/d$. Strang propagation splits $G_m = G_B + (G_m - G_B)$ symmetrically within each free segment. Substep counts are 8, 16, 32, 64 and 128; they refine the propagator and do not refine a physical finite-width pulse.

CCE orders 1–4 and pair cutoffs 5, 10, 20 and 40 angstrom are compared with exact four-spin dynamics. Bath sizes 2, 4, 6 and 8 and time-grid sizes 33, 65, 129 and 257 compare adjacent nested levels; their initial levels are inconclusive because no preceding comparator exists. The default curve grid has 33 ages over 0–0.4 ms. Three pulse choices (0, 1, 4 ideal CPMG pulses) give the exact count

$$3\,[2 + 5 + 4 + 4 + 4 + 4] = 69$$

solver calls. There are 66 diagnostics because a single independent-solver row uses two calls. Final tolerances are 10^{-10} for independent solvers, 10^{-4} for Strang/time-grid controls and 10^{-3} for CCE, cutoff and bath expansion. Failed coarser levels are retained; final rows pass. The time-grid metric is trapezoidal mean $|L|$, not all extrema or a fitted lifetime.

The weak outer-coupling construction is a deliberate numerical control. Eight-spin convergence under these recipes says nothing about a dense material lattice, sample disorder ensemble or another time interval. No pulse-shape, pulse-error, transverse central-spin relaxation or phonon-model convergence is supplied. For profiles labeled `prediction_purpose: material_prediction`, the new gate checks certificate/profile hashes, all five refinement axes, error/tolerance, validated times, nonhypothetical evidence and independent physical-model review. It is a declared-evidence guard, not an automated replacement for scientific review; no certificate for a real material is issued.

S14 Native simulator contracts and larger-study accounting

SeQUeNCe uses its `Timeline`, native events/processes and joint density manager. QuNetSim uses native `Host`/`Network` connections, quantum packets, ownership transfer and ACKs. A custom QuTiP density backend implements the local operations/channel action needed by this two-node protocol; its joint-state budget is four qubits. It preserves tensor ordering, releases consumed qubits and rejects unsupported measurement calls. Stock EQSN, general large-network operations and native physical timing are not validated by this backend. The simulator reference is DiAdamo et al., IEEE Transactions on Quantum Engineering 2 (2021).

Each heralded attempt prepares an ideal Bell pair and transfers one half between the QuNetSim hosts even if later readout loses it. Both retrievals apply the modeled signal or background branch, or erase the pair; all qubits are released after the attempt. The SeQUeNCe arm schedules loading/retrieval and resets the corresponding joint state after each cycle. An independent tensor-matrix-unit reference checks the local and joint evolution. The two arms use different channel application routes and the same external event outcome inputs.

S14.1 Parameters, RNG and quality semantics

Sixteen combinations use $r_0 \in \{0, 5000\}$ s^{-1}, ages $\{0, 0.001, 0.01, 0.03\}$ s and herald probabilities $\{0.25, 0.75\}$. Constant rates are population 30, pure dephasing 50 and erasure 12 s^{-1}, with equilibrium population 0.2 and detuning 7 Hz. Writing/reading use efficiencies 0.8/0.7, depolarization 0.01/0.02, durations 100/200 μs and specified 4.2 K conditions. Readout uses a 10 μs window, background population 0.5 and age coefficient 50 000 s^{-2}. The 40 ms attempt interval exceeds the complete cycle at every tested age. No overlap, routing, swapping, fiber propagation or contention is simulated.

Master seed 20261010 is combined with each study seed and stream name through SHA-256; the first eight digest bytes interpreted little-endian seed NumPy's PCG64 generator. Three stream names are `herald`, `left_readout` and `right_readout`. The resulting exact uniform arrays are shared by both simulator arms and retained. Parameter cases share streams intentionally, so they are correlated; seeds, not all individual attempts or parameter rows, are independent Monte Carlo clusters.

The signal/noise/loss thresholds are η, $\eta + q$ and one. Internal branch labels diagnose false-positive handling but do not decide whether an individual pair meets a quality requirement. Quality-qualified counts instead use the common model ensemble Bell fidelity $F_B \geq 0.8$ and divide by modeled attempt time. Background can increase raw acceptance while reducing fidelity. Figure points show all eight seed outcomes without inferential intervals or claims of speed superiority.

S14.2 Pilot, larger study and limitations

The pilot has seeds 1101–1103 and 32 attempts per arm: $16 \times 3 \times 2 = 96$ arm runs and 3072 attempts. After passing instrument/convergence/pilot prerequisites, seeds 2101–2108 and 512 attempts produce 256 arm runs and 131 072 attempts. All 128 larger paired comparisons have identical branch counts and logical hashes; maximum mean-density error is 2.465×10^{-15}. Native event and quantum-transfer counts are 98 344 and 32 808. No selected arm fails. Across pilot and larger stages there are 352 arm runs, 176 paired comparisons and 134 144 attempts.

Both native stages run in separate child processes; QuNetSim's global network is never restarted after a completed stage in the same interpreter. QuNetSim's wall-clock queues do not provide the physical time used by this study. Storage age and event timestamps come from the supplied common logical trace, so equal hashes demonstrate trace adherence rather than equal native latency. The scientific scope is thus state/instrument consistency and native two-node transport under common external inputs, not physical network-timing equivalence or larger-network scalability.

S15 Current replay, claim mapping and manuscript build

The extension's actual source base is git commit `985c8fe`; the source changes were uncommitted when the selected computations ran. Their content identities, rather than that base alone, identify the executed implementation: lab hash

`3852565376c2a529bb33bc9394cf96061eeee147e19b4f4bbc2da724cf224f5a`,

and evaluation hash

`ecc4ea6d77ca774a7f9cd3ed6f91285a5d5632a041ecdaa79ff829a5464cc373`.

The published result commit is 9358173. Manuscript/plotting scripts are outside the lab/evaluation fingerprints; the reporting generator verifies that the currently delivered source matches these content identities. Historical Bradley, repair and original-paper executions retain their own fingerprints and are not relabeled as executions at the new commit.

The extension environment is Linux/Python 3.12.14, NumPy 2.3.5, SciPy 1.16.3, Matplotlib 3.10.8, PyCCE 1.1.1, SeQUeNCe 1.2.0, QuNetSim 0.1.2, QuTiP 5.3.1 and QuTiP-QIP 0.4.2. Its 126-distribution snapshot is `evaluation/config/validation_extension/requirements-tested.txt`. Install it in a separate Python 3.12 environment from older snapshots. PyCCE compatibility uses the retained SciPy utility and NumPy Unicode alias documented in Section S10; installed dependency source is not patched.

After installation, use a fresh output directory:

```
export OMP_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 MKL_NUM_THREADS=1
export WQ_BLOCK_EXTERNAL_NETWORK=1
export PYTHONPATH="$PWD/scripts/network_guard:$PWD/lab"
python scripts/run_validation_extension.py -count
python scripts/run_validation_extension.py -out reproductions/my-extension
python scripts/verify_validation_extension.py
-reference evaluation/results/validation_extension/selected
-actual reproductions/my-extension
```

The last three lines form one command. `-resume` requires matching configuration and code fingerprints. Every native run JSON and RNG array is retained in the two `raw_runs_and_traces.zip` files; their uncompressed duplicates are ignored by git. A fresh offline replay checked 359 JSON records and 11 exact RNG arrays, with equal scientific contents at absolute/relative tolerance 10^{-10}. Runtime, timestamps/platform provenance and ZIP-container bytes are excluded, but per-archive integrity and each archived scientific record are checked.

The current environment runs 34 evaluation tests plus 75 lab tests: 107 pass and 2 optional PySCF/MaxwellBloch tests are skipped. Eleven extension tests cover noise/efficiency distinctions, false-positive limits, double counting, independent noncommuting propagation, certificate identity/time limits, explicit PyCCE pairs/Zeeman conventions, independent RNG streams and joint-state preservation/consumption in the custom backend. Historical full optional tests are a separate snapshot. The current system regression explicitly permits changed source hashes while comparing all four full scientific payloads; its strict pending physical case remains blocked. All replay/test checks are author-operated and not independent scientific review.

`scripts/document_article_update.py` validates retained numerical values, status limitations, raw-data/result/archive hashes, source fingerprints, run counts and test logs, then generates two LaTeX tables, number macros and `evidence/article_update_claim_map.json`. The original ten-claim and system-evaluation maps remain historical. Extension figures are generated in English by `scripts/plot_validation_extension.py -language en` and `scripts/plot_article_bradley.py`; earlier Swedish figures remain unchanged. `make pdf` builds the article and this Supplement from included assets. TeX compilation and complete rendered-page inspection are recorded separately in the current manuscript-release evidence. These checks establish a complete report of the executed study; no new experimental data, independent material review or journal acceptance is implied.